Machine Learning

→ Developing algorithm that allows a computer to improve its performance through experience to make smart and accurate decisions.

Traditional Programing And machine Learning:-

→ Procedural programing Like using loops etc.

→ some approach towards oop.

→ Data collection.

1. Data collection:-

inputs → from keggle / web scrapping.

impages / videos.

→ text, images, voice

- ↓ labelled Data
- ↓ unlabelled Data
- → L/U Data.

attributes → labelled.

S.no	Name	DOB
1	MRK	123
2	SRK	456
3	ARK	789
4	MAK	032.

2. Learning

→ Analysis data

→ Find patterns

ex

$2 \times 1 = 2$

$2 \times 2 = 4$

$2 \times 3 = ? \rightarrow 6.$

3. Predictions:-

→ make predications. / find unknown results.

$2 \times 3 = ?$ 6 → prediction.

4. Decision Making:-

Take decisions if the result is good or not means correct or not.

Types of Meachine Learning:-

1. Traditional Teaching (Superviced)

I/P + labels ⇒ prediction.

Classification:-
→ Categories checking

Regreration:-
→ predicting a number.

2. Self Discovery (Unsuperviced)

→ unlabelled Data.

Clustering:-
make groups of data.

images / live streaming (CCTV)

3. Semi supesviced Data:-

Combination of superviced and Unsuperviced Data.

Label + UnLabel ⟶ Learning process.

→ started from Labelled data.

e.g Red like apple
Yellow like banana } → automatically label.

4. Reinforcement Learning:-

→ Rewards and panelties.

THE 11 STEPS TO BUILD AND DEPLOY YOUR AI APP

Step 1: Define the problem.
→ Aim → Question Raised (WHY!)

Step 2:- Data collection And Gathering
→ images, text, Videos etc.

Step 3:- Data preprocessing:-
→ most important step.
→ Removing outliers.
→ Tiding Data.

Step 4:- Choose a model:-
Classification, Regressions and clustering.

Step 5:- Splitting the Data:-
→ Training and Testing splitting.
→ 80/20 → Testing
↓
Training

Step 6:- Evaluating the Model:-
R^2, RMSE, MSE.

Step 7:- Hyperparameter Tuning:-
→ Tune the Errors

Step 8:- Cross Validation (K-Fold)
→ Different folds.

Step 09: Model Finalizations
→ Finish.
→ Lock in a final model.

Step 10:- Deployment
→ Deploy on machine → Flask - etc.

Step 11:- MLOps (Monitor & Update)
→ this done in companys